Section J
Differentiation ($f'(x) = ?$)

1. $[\cosh(x^2+4x)](2x+4)$

2. $7\cosh(7x)$

3. $[\sinh(x^4+5x+7)](4x^3+5)$

4. $[\operatorname{sech}^2(\sinh 2t)](\cosh(2t))(2)$

5. $[\operatorname{sech}^2(x^2+5)](2x)$

6. $\dfrac{(\cosh x)(x) - (\sinh x)(1)}{x^2}$

7. $[\operatorname{sech}^2(2y-y^3)](2-3y^2)$

8. $[3\cosh(x^2)](2x) + 2\sinh x + 0$

9. $(3\cosh(4x))(4) - (2\sinh(5x))(5)$

10. $[\sinh(\tanh x)](\operatorname{sech}^2 x)$

11. $[\sinh(\frac{1}{\sqrt{x}})](-\frac{1}{2})(x^{-3/2})$

12. $\operatorname{sech}^2(\frac{1}{x})(-\frac{1}{x^2})$

13. $-\cosh(-t)$

14. $\dfrac{(\sinh x)(1+x) - \cosh x}{(1+x)^2}$

15. $2[4\cosh(3x)](\sinh(3x))(3)$

16. $2[\sinh(x^4)](\cosh(x^4))(4x^3)$

17. $\frac{d}{dx}(\cosh(2x)) = 2\sinh(2x)$

18. $f'(\theta) = \dfrac{2}{\sqrt{1-(2\theta)^2}}$

19. $\dfrac{-2x}{\sqrt{1-(x^2)^2}}$

20. $\arcsin x + (x)\left(\dfrac{1}{\sqrt{1-x^2}}\right)$

21. $\left(\dfrac{-2}{x^2}\right)\left(\dfrac{1}{\sqrt{1-(\frac{2}{x})^2}}\right)$

22. $\dfrac{2}{\sqrt{1-(2y-3)^2}}$

23. $2\arctan x + (2x)\left(\dfrac{1}{1+x^2}\right)$

24. $\dfrac{5}{1+(5x)^2}$

25. $\dfrac{\cos t}{\sqrt{1-(\sin t)^2}} = \dfrac{\cos t}{\cos t} = 1$

26. $\dfrac{3}{1+(3x-4)^2}$

27. $\dfrac{-\frac{1}{4}}{\sqrt{1-(\frac{x}{4})^2}}$